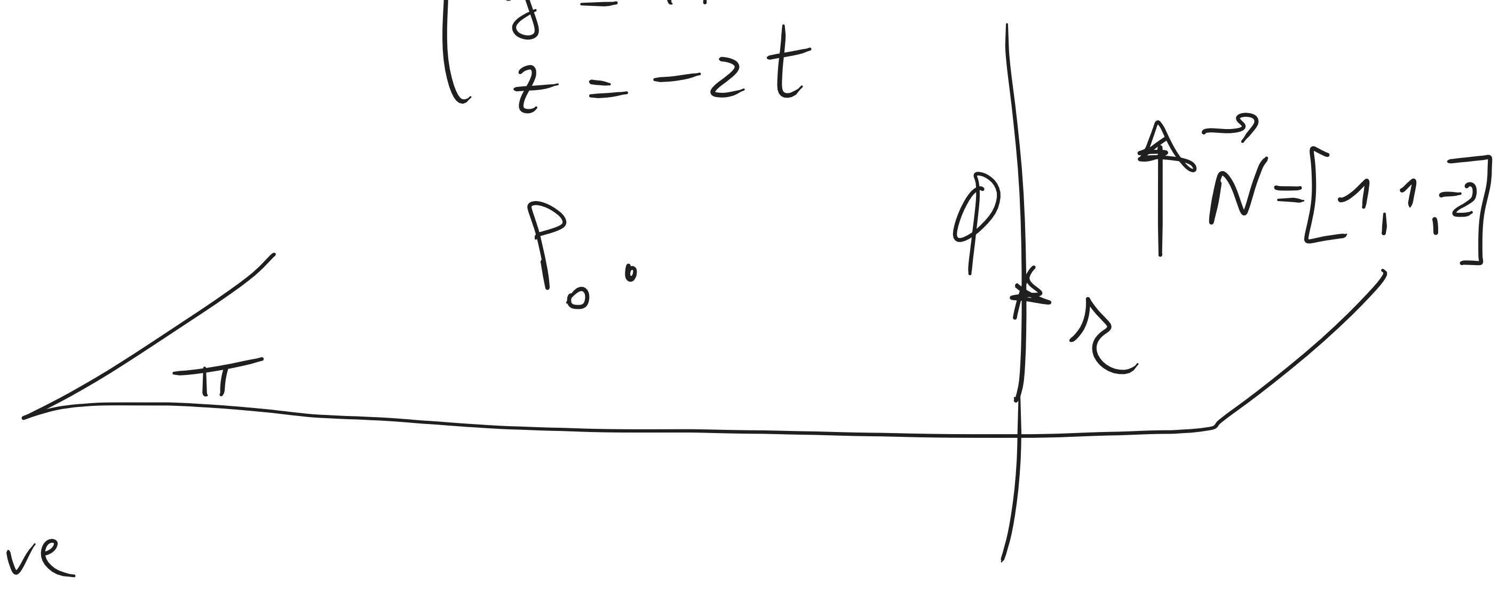
Es. $P_0 = [1, -1, 2]$ $\quad r: \begin{cases} x = t \\ y = 1 + t \\ z = -2t \end{cases}$

Calcolare $\mathrm{dist}(P_0, r)$



$P_0 \notin r$

$r: [t, 1+t, -2t]$

$\mathrm{dist}(P_0, r) = \mathrm{dist}(P_0, Q)$, dove

$\underbrace{Q = r \cap \pi}$ $\quad \pi: x + y - 2z + 4 = 0$ Il punto Q corrisponde a quel valore di t per cui: $t + (1+t) - 2(-2t) + 4 = 0$ $\quad 6t = -5$

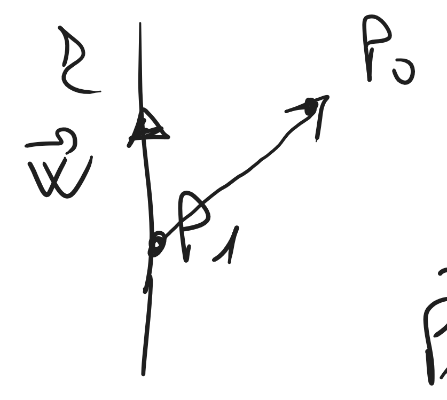
$t = -\frac{5}{6}$. $\quad Q = \left[-\frac{5}{6}, \frac{1}{6}, \frac{5}{3}\right]$ $\quad P_0 = [1, -1, 2]$

$$\mathrm{dist}(P_0, r) = \mathrm{dist}(P_0, Q) = \sqrt{\left(\frac{11}{6}\right)^2 + \left(\frac{7}{6}\right)^2 + \left(\frac{2}{6}\right)^2} = \frac{\sqrt{174}}{6} = \frac{\sqrt{29}}{\sqrt{6}}$$

$r: [t, 1+t, -2t]$ $\quad P_0 = [1, -1, 2]$ $\quad \underbrace{\vec{w} \text{ versore} \parallel r}$

$P_1 \in r$

$\mathrm{dist}(P_0, r) = |\underbrace{\vec{w} \wedge (\overrightarrow{P_0 - P_1})}|$



$P_1 = [0, 1, 0]$

$\vec{w} = \mathrm{vers}([1, 1, -2])$

$= \frac{1}{\sqrt{6}} [1, 1, -2]$

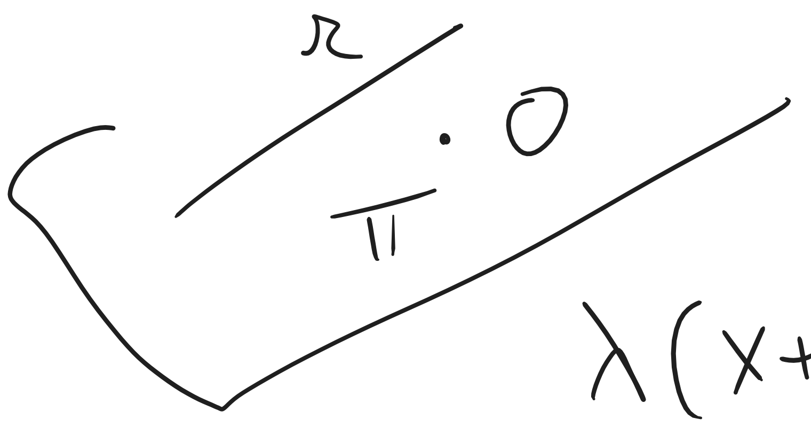
$\overrightarrow{P_0 - P_1} = [1, -2, 2]$

$$\vec{w} \wedge (\overrightarrow{P_0 - P_1}) = \frac{1}{\sqrt{6}} \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 1 & -2 \\ 1 & -2 & 2 \end{vmatrix} = \frac{1}{\sqrt{6}} [-2, -4, -3]$$

$$\mathrm{dist}(P_0, r) = |\vec{w} \wedge (\overrightarrow{P_0 - P_1})| = \left| \frac{1}{\sqrt{6}} [-2, -4, -3] \right| = \frac{\sqrt{4 + 16 + 9}}{\sqrt{6}} = \frac{\sqrt{24}}{\sqrt{6}}$$

Es. $r: \begin{cases} x + y + z = 0 \\ 3x - y - z - 1 = 0 \end{cases}$ $\quad P_0 = O = [0, 0, 0]$

Determinare l'eq. del piano π che contiene r e O



π appartiene al fascio generato da r e quindi π è del tipo:

$$\lambda(x + y + z) + \mu(3x - y - z - 1) = 0$$

Devo imporre passaggio per O:

$$\lambda(0 + 0 + 0) + \mu(0 - 0 - 0 - 1) = 0$$

$$0 \cdot \lambda - \mu = 0 \rightarrow \boxed{\mu = 0} \qquad \underline{\lambda = 1}$$

$\pi: x + y + z = 0$

Es. Determinare l'eq. del piano π che passa per $P_0 = [1, 2, 3]$ e contiene $\underline{\text{l'asse } y}$

$$\boxed{\pi : 3x - z = 0}$$

verificare!